

Trigonometry (10 th std)

Choose the Appropriate Answer

1. The value of $(\sin^2 \theta + 1) / (1 + \tan^2 \theta)$ is equal to

- (a) $\tan^2 \theta$ (b) 1 (c) $\cot^2 \theta$ (d) 0

Ans: (b) 1

$$\sin^2 \theta + 1 / 1 + \tan^2 \theta = \sin^2 \theta + 1 \sec^2 \theta = \sin^2 \theta + \cos^2 \theta = 1$$

2. $\tan \theta \operatorname{cosec}^2 \theta - \tan \theta$ is equal to

- (a) $\sec \theta$ (b) $\cot^2 \theta$ (c) $\sin \theta$ (d) $\cot \theta$

Ans: (d) $\cot \theta$

$$\tan \theta \operatorname{cosec}^2 \theta - \tan \theta = \tan \theta (\operatorname{cosec}^2 \theta - 1) = \tan \theta \times \cot^2 \theta = 1 \cot \theta \times \cot^2 \theta = \cot \theta$$

3. If $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\cos \alpha + \sec \alpha)^2 = k + \tan^2 \alpha + \cot^2 \alpha$, then the value of k is equal to

- (a) 9 (b) 7 (c) 5 (d) 3

Ans: (b) 7

$$\begin{aligned} & (\sin \alpha + \operatorname{cosec} \alpha)^2 + (\cos \alpha + \sec \alpha)^2 \\ &= \sin^2 \alpha + \operatorname{cosec}^2 \alpha + 2 \sin \alpha \operatorname{cosec} \alpha + \cos^2 \alpha + \sec^2 \alpha + 2 \cos \alpha \sec \alpha \\ &= 1 + \operatorname{cosec}^2 \alpha + 2 + \sec^2 \alpha + 2 \\ &= 1 + \cot^2 \alpha + 1 + 2 + \tan^2 \alpha + 1 + 2 = 7 + \tan^2 \alpha + \cot^2 \alpha \Rightarrow k = 7 \end{aligned}$$

4. If $\sin \theta + \cos \theta = a$, and $\sec \theta + \operatorname{cosec} \theta = b$, then the value of $b(a^2 - 1)$ is equal to

- (a) 2 a (b) 3 a (c) 0 (d) 2 ab

Ans: (a) 2 a

$$\begin{aligned} b(a^2 - 1) &= (\sec \theta + \operatorname{cosec} \theta) [(\sin \theta + \cos \theta)^2 - 1] \\ &= 1 \cos \theta + 1 \sin \theta [\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1] \\ &= \left(\frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} \right) [1 + 2 \sin \theta \cos \theta - 1] = \left(\frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} \right) \times 2 \sin \theta \cos \theta \\ &= 2 (\sin \theta + \cos \theta) = 2 a \end{aligned}$$

5. If $5x = \sec \theta$ and $5x = \tan \theta$, then $x^2 - 1/x^2$ is equal to

- (a) 25 (b) 1 / 25 (c) 5 (d) 1

Ans: (b) 1 / 25

$$\begin{aligned} 5x &= \sec \theta \Rightarrow x = \left(\frac{\sec \theta}{5} \right) \\ x^2 &= \frac{\sec^2 \theta}{25} \Rightarrow \frac{1}{x} = \frac{\tan \theta}{5} \Rightarrow \frac{1}{x^2} = \frac{\tan^2 \theta}{25} \\ x^2 - \frac{1}{x^2} &= \frac{\sec^2 \theta}{25} - \frac{\tan^2 \theta}{25} = \frac{\sec^2 \theta - \tan^2 \theta}{25} = \frac{1}{25} \end{aligned}$$

6. If $\sin \theta = \cos \theta$, then $2 \tan^2 \theta + \sin^2 \theta - 1$ is equal to

- (a) $-3/2$ (b) $3/2$ (c) $2/3$ (d) $-2/3$

Ans: (b) $3/2$

7. If $x = a \tan \theta$ and $y = b \sec \theta$ then

- (a) $y^2 b^2 - x^2 a^2 = 1$ (b) $x^2 a^2 - y^2 b^2 = 1$
(c) $x^2 a^2 - y^2 b^2 = 0$ (d) $y^2 b^2 - x^2 a^2 = 0$

Ans: (a) $y^2 b^2 - x^2 a^2 = 1$

$$y^2 / b^2 - x^2 / a^2 = \sec^2 \theta - \tan^2 \theta = 1$$

8. $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)$ is equal to

- (a) 0 (b) 1 (c) 2 (d) -1

Ans: (c) 2

$$(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)$$

$$= \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right) \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) = \left(\frac{\cos \theta + \sin \theta + 1}{\cos \theta}\right) \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right)$$

$$= \frac{(\sin \theta + \cos \theta)^2 - 1}{\sin \theta \cos \theta} = \frac{\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} = \frac{1 + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} = \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} = 2$$

9. $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$ then $p^2 - q^2$ is equal to

- (a) $a^2 - b^2$ (b) $b^2 - a^2$ (c) $a^2 + b^2$ (d) $b - a$

Ans: (b) $b^2 - a^2$

$$\begin{aligned} p^2 - q^2 &= a^2 \cot^2 \theta + a^2 \cot^2 \theta + 2ab \cot \theta \operatorname{cosec} \theta - (b^2 \cot^2 \theta + a^2 \operatorname{cosec}^2 \theta + 2ab \cot \theta \operatorname{cosec} \theta) \\ &= (a^2 - b^2) \cot^2 \theta + (b^2 - a^2) \operatorname{cosec}^2 \theta = (a^2 - b^2) (\operatorname{cosec}^2 \theta - 1) + (b^2 - a^2) (\operatorname{cosec}^2 \theta) \\ &= (a^2 - b^2) \operatorname{cosec}^2 \theta - (a^2 - b^2) - (a^2 - b^2) \operatorname{cosec}^2 \theta = b^2 - a^2 \end{aligned}$$

10. If the ratio of the height of a tower and the length of its shadow is $\sqrt{3}:1$, then the angle of elevation of the sun has a measure

- (a) 45° (b) 30° (c) 90° (d) 60°

Ans: (d) 60°

11. In $\triangle ABC$, right-angled at B, $AB = 24$ cm, $BC = 7$ cm. The value of $\tan C$ is:

- (a) $12/7$ (b) $24/7$ (c) $20/7$ (d) $7/24$

Ans: (b) $24/7$

$$\tan C = \text{Opposite side} / \text{Adjacent side} = 24/7$$

12. $(\sin 30^\circ + \cos 60^\circ) - (\sin 60^\circ + \cos 30^\circ)$ is equal to:

- (a) 0 (b) $1 + 2\sqrt{3}$ (c) $1 - \sqrt{3}$ (d) $1 + \sqrt{3}$

Ans: (c) $1 - \sqrt{3}$

$\sin 30^\circ = \frac{1}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$ and $\cos 60^\circ = \frac{1}{2}$

Putting these values, we get:

$$(\frac{1}{2} + \frac{1}{2}) - (\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}) = 1 - [(2\sqrt{3}) / 2] = 1 - \sqrt{3}$$

13. The value of $\tan 60^\circ / \cot 30^\circ$ is equal to:

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (b) 1

$$\tan 60^\circ / \cot 30^\circ = \sqrt{3} / \frac{1}{\sqrt{3}} = 1$$

14. $1 - \cos^2 A$ is equal to:

- (a) $\sin^2 A$ (b) $\tan^2 A$ (c) $1 - \sin^2 A$ (d) $\sec^2 A$

Ans: (a) $\sin^2 A$

15. $\sin(90^\circ - A)$ and $\cos A$ are:

- (a) Different (b) Same (c) Not related (d) None of the above

Ans: (b) Same

16. If $\cos X = \frac{2}{3}$ then $\tan X$ is equal to:

- (a) $5/2$ (b) $\sqrt{5/2}$ (c) $\sqrt{5}/2$ (d) $2/\sqrt{5}$

Ans: (c) $\sqrt{5}/2$

$$1 + \tan^2 X = \sec^2 X \text{ and } \sec X = 1 / \cos X = 1 / (\frac{2}{3}) = \frac{3}{2}$$

$$1 + \tan^2 X = (\frac{3}{2})^2 = 9/4 \Rightarrow \tan^2 X = (9/4) - 1 = 5/4 \Rightarrow \tan X = \sqrt{5}/2$$

17. If $\cos X = a/b$, then $\sin X$ is equal to:

- (a) $(b^2 - a^2) / b$ (b) $(b - a) / b$ (c) $\sqrt{(b^2 - a^2)} / b$ (d) $\sqrt{(b - a)} / b$

Ans: (c) $\sqrt{(b^2 - a^2)} / b$

18. The value of $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ is:

- (a) 0 (b) 1 (c) 2 (d) 4

Ans: (b) 1

19. $2 \tan 30^\circ / (1 + \tan^2 30^\circ) =$

- (a) $\sin 60^\circ$ (b) $\cos 60^\circ$ (c) $\tan 60^\circ$ (d) $\sin 30^\circ$

Ans: (a) $\sin 60^\circ$

20. $\sin 2A = 2 \sin A$ is true when $A =$

- (a) 30° (b) 45° (c) 0° (d) 60°

Ans: (c) 0°

21. The value of $(\sin 45^\circ + \cos 45^\circ)$ is

- (a) $1/\sqrt{2}$ (b) $\sqrt{2}$ (c) $\sqrt{3}/2$ (d) 1

Ans: (b) $\sqrt{2}$

22. If $\sin A = 1/2$, then the value of $\cot A$ is
 (a) $\sqrt{3}$ (b) $1/\sqrt{3}$ (c) $\sqrt{3}/2$ (d) 1

Ans: (a) $\sqrt{3}$

23. If ΔABC is right angled at C, then the value of $\cos (A+B)$ is
 (a) 0 (b) 1 (c) $1/2$ (d) $\sqrt{3}/2$

Ans: (a) 0

$$\begin{aligned} \angle A + \angle B + \angle C &= 180^\circ \Rightarrow \angle A + \angle B + 90^\circ = 180^\circ (\because \angle C = 90^\circ) \\ \angle A + \angle B &= 90^\circ \Rightarrow \text{Now, } \cos (A+B) = \cos 90^\circ = 0 \end{aligned}$$

24. The value of $(\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ)$ is
 (a) 0 (b) 1 (c) 2 (d) $1/2$

Ans: (b) 1

$$\begin{aligned} \tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ &= [\tan 1^\circ \tan 2^\circ \dots \tan 44^\circ] \tan 45^\circ [\tan (90^\circ - 44^\circ) \tan (90^\circ - 43^\circ) \dots \tan (90^\circ - 1^\circ)] \\ &= [\tan 1^\circ \tan 2^\circ \dots \tan 44^\circ] [\cot 44^\circ \cot 43^\circ \dots \cot 1^\circ] \times [\tan 45^\circ] \\ &= [(\tan 1^\circ \times \cot 1^\circ) (\tan 2^\circ \times \cot 2^\circ) \dots (\tan 44^\circ \times \cot 44^\circ)] \times [\tan 45^\circ] \\ &= 1 \times 1 \times 1 \times 1 \times \dots \times 1 \quad \{\text{since } \tan A \times \cot A = 1 \text{ and } \tan 45^\circ = 1\} = 1 \end{aligned}$$

25. The value of the expression $[\operatorname{cosec} (75^\circ + \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot (35^\circ - \theta)]$ is
 (a) -1 (b) 0 (c) 1 (d) $3/2$

Ans: (b) 0

$$\begin{aligned} &[\operatorname{Cosec} (75^\circ + \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot (35^\circ - \theta)] \\ &= \operatorname{cosec} [90^\circ - (15^\circ - \theta)] - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot [90^\circ - (55^\circ + \theta)] \\ &\text{We know that } \operatorname{cosec} (90^\circ - \theta) = \sec \theta \text{ and } \cot (90^\circ - \theta) = \tan \theta. \\ &= \sec (15^\circ - \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \tan (55^\circ + \theta) = 0 \end{aligned}$$

26. If $\cos(\alpha + \beta) = 0$, then $\sin(\alpha - \beta)$ can be reduced to
 (a) $\cos \beta$ (b) $\cos 2\beta$ (c) $\sin \alpha$ (d) $\sin 2\alpha$

Ans: (b) $\cos 2\beta$

27. If $\sin A + \sin^2 A = 1$, then the value of the expression $(\cos^2 A + \cos^4 A)$ is
 (a) 1 (b) $1/2$ (c) 2 (d) 3

Ans: (a) 1

28. If $\cos 9\alpha = \sin \alpha$ and $9\alpha < 90^\circ$, then the value of $\tan 5\alpha$ is
 (a) $1/\sqrt{3}$ (b) $\sqrt{3}$ (c) 1 (d) 0

Ans: (c) 1

$$\begin{aligned} \cos 9\alpha &= \cos (90^\circ - \alpha) \Rightarrow 9\alpha = 90^\circ - \alpha \\ 9\alpha + \alpha &= 90^\circ \Rightarrow 10\alpha = 90^\circ \Rightarrow \alpha = 9^\circ \\ \tan 5\alpha &= \tan (5 \times 9^\circ) = \tan 45^\circ = 1 \end{aligned}$$

29. $\left[\frac{\sin^2 22^\circ + \sin^2 68^\circ}{\cos^2 22^\circ + \cos^2 68^\circ} + \sin^2 63^\circ + \cos 63^\circ \sin 27^\circ \right]$
 (a) 3 (b) 2 (c) 1 (d) 0

Ans: (b) 2

$$\begin{aligned} & \left[\frac{\sin^2 22^\circ + \sin^2 68^\circ}{\cos^2 22^\circ + \cos^2 68^\circ} + \sin^2 63^\circ + \cos 63^\circ \sin 27^\circ \right] \\ &= \left[\frac{\sin^2 22^\circ + \sin^2 (90 - 22^\circ)}{\cos^2 22^\circ + \cos^2 (90 - 22^\circ)} + \sin^2 63^\circ + \cos 63^\circ \sin (90 - 63^\circ) \right] \\ &= \left[\frac{\sin^2 22^\circ + \sin^2 (90 - 22^\circ)}{\cos^2 22^\circ + \cos^2 (90 - 22^\circ)} + \sin^2 63^\circ + \cos 63^\circ \cos 63^\circ \right] \\ &= \frac{\sin^2 22^\circ + \cos^2 22^\circ}{\cos^2 22^\circ + \sin^2 22^\circ} + \sin^2 63^\circ + \cos^2 63^\circ = 1 + 1 = 2 \end{aligned}$$

30. The value of the expression $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$ is
 (a) 0 (b) 3 (c) 2 (d) 1

Ans: (d) 1

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \quad (\text{Taking cube on both sides}) \\ (\sin^2 \theta + \cos^2 \theta)^3 &= 1^3 \quad (\sin^2 \theta)^3 + (\cos^2 \theta)^3 + 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) = 1 \\ \sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta &= 1 \end{aligned}$$

31. If $\sin (A + B) = 3\sqrt{2}$ and $\tan (A - B) = 1$. What are the values of A and B?
 (a) 37, 54 (b) 35.7, 40.7 (c) 50, 10 (d) 52.5, 7.5

Ans: (d) 52.5, 7.5

$$\begin{array}{l} \text{Adding equation (1) and (2)} \\ A + B = 60 \\ + A - B = 45 \\ \hline 2A = 105 \end{array} \quad \boxed{A = 52.5, B = 7.5}$$

32. If $\cos \theta = 34$ then value of $\cos 2\theta$ is _____
 (a) 16 (b) 14 (c) 18 (d) 38

Ans: (c) 18

$$\cos 2\theta = 2\cos^2 \theta - 1 \Rightarrow \cos 2\theta = 2(34)^2 - 1 = 18$$

33. If $\sin A = 8 / 17$, what will be the value of $\cos A \sec A$?
 (a) 2 (b) -1 (c) 1 (d) 0

Ans: (c) 1

34. The value of each of the trigonometric ratios of an angle depends on the size of the triangle and does not depend on the angle.

(a) True (b) False (c) can't determine (d) None of these

Ans: (b) False

35. If $\tan \alpha = \sqrt{3}$ and $\operatorname{cosec} \beta = 1$, then the value of $\alpha - \beta$?

- (a) -30° (b) 30° (c) 90° (d) 60°

Ans: (a) -30°

$$\begin{array}{ll} \tan \alpha = \sqrt{3} \text{ and } \tan 60^\circ = \sqrt{3} & \therefore \alpha = 60^\circ \\ \operatorname{Cosec} \beta = 1 \text{ and } \operatorname{cosec} 90^\circ = 1 & \therefore \beta = 90^\circ \\ \alpha - \beta = 60 - 90 = -30^\circ & \end{array}$$

36. In triangle ABC, right angled at C, then the value of $\operatorname{cosec} (A + B)$ is ____.

- (a) 2 (b) 0 (c) 1 (d) ∞

Ans: (c) 1

37. If $\tan \theta = 3/4$ then the value of $\sin \theta$ is _____

- (a) $3/5$ (b) $4/4$ (c) $3/4$ (d) $-3/5$

Ans: (a) $3/5$

38. What is the value of $\sin 30^\circ \cos 15^\circ + \cos 30^\circ \sin 15^\circ$?

- (a) 12 (b) 0 (c) 1 (d) $1/\sqrt{2}$

Ans: (d) $1/\sqrt{2}$

39. What is the value of $\cos A \sec A + \sin A \operatorname{cosec} A - \tan A \cot A$?

- (a) 0 (b) 2 (c) 1 (d) 3

Ans: (c) 1

40. In a right-angled triangle, the trigonometric function that is equal to the ratio of the side opposite a given angle to the hypotenuse is called cosine.

- (a) False (b) True (c) Can't determine (d) None of these

Ans: (a) False

41. Trigonometric ratios are only applicable to which kind of triangles?

- (a) Right-angled triangles (b) Any type of triangles
(c) Acute angled triangles (d) Obtuse angled triangles

Ans: (a) Right-angled triangles

42. Trigonometric ratios are _____

- (a) sine, cosine and cotangent
(b) sine, tangent, cotangent and secant
(c) sine, cosine, tangent, cotangent, secant and cosecant
(d) tangent, cotangent and secant

Ans: (c) sine, cosine, tangent, cotangent, secant and cosecant

43. What is the inverse of cosecant?

- (a) Cosine (b) Secant (c) Sine (d) Tangent

Ans: (c) Sine

44. $\sin \theta / \cos \theta$ equals to _____

- (a) $\tan \theta$ (b) $\cot \theta$ (c) $\operatorname{cosec} \theta$ (d) $\sec \theta$

Ans: (a) $\tan \theta$

45. Who is the 'Father of Trigonometry'?

- (a) Hipparchus (b) Euclid (c) Aristotle (d) Archimedes

Ans: (a) Hipparchus

46. The product of $\sin \theta$ and $\operatorname{Cosec} \theta$ is 1.

- (a) True (b) False (c) Can't determine (d) None of these

Ans: (a) True

47. What is the cotangent of an angle θ if the side adjacent to θ is 8 units and the side opposite to θ is 3 units?

- (a) $3/8$ (b) $8/3$ (c) $4/3$ (d) $3/4$

Ans: (b) $8/3$

48. Choose the correct reciprocal ratios.

- (a) $\tan \theta$, $\sec \theta$ (b) $\operatorname{Cosec} \theta$, $\sec \theta$ (c) $\sec \theta$, $\sin \theta$ (d) $\tan \theta$, $\cot \theta$

Ans: (d) $\tan \theta$, $\cot \theta$

49. The basic trigonometric ratios are _____

- (a) sine (b) tangent (c) sine, cosine and tangent (d) cosine

Ans: (c) sine, cosine and tangent

50. Length of hypotenuse / Length of opposite to angle A is _____

- (a) sine of angle A (b) cosec of angle A
(c) tan of angle A (d) cot of angle A

Ans: (b) cosec of angle A

51. If $\sin A = 3/5$, then find $\tan A$.

- (a) $3/4$ (b) $3/5$ (c) $5/3$ (d) $4/3$

Ans: (a) $3/4$

52. $\cos A / \sin A =$ _____

- (a) $\tan A$ (b) $\sin A$ (c) $\cot A$ (d) $\sec A$

Ans: (c) $\cot A$

53. If $\sin B = 3/5$, then find $\sec B$.

- (a) $3/4$ (b) $4/5$ (c) $5/4$ (d) $5/3$

Answer: (c) $5/4$

54. What is the value of $\cos^2 \theta - \sin^2 \theta$ if the length of the opposite side is 20 units and the length of the hypotenuse is 29 units?

- (a) -41841 (b) -41840 (c) 41841 (d) 41840

Ans: (a) -41841

$$(\text{Adjacent side})^2 = (\text{Hypotenuse})^2 - (\text{Opposite side})^2 \Rightarrow \text{Adjacent side} = \sqrt{441} = 21$$

$$\cos \theta = \frac{\text{Length of the adjacent side}}{\text{Length of the hypotenuse}} = \frac{21}{29}, \quad \sin \theta = \frac{\text{Length of the opposite side}}{\text{Length of the hypotenuse}} = \frac{20}{29}$$

$$\cos^2 \theta - \sin^2 \theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = -\frac{41841}{841}$$

55. If the length of the side opposite to angle A is 15 units and the length of the hypotenuse is 17 units then the length of the side adjacent to angle A is ____
 (a) 8 units (b) 7 units (c) 4 units (d) 5 units

Ans: (a) 8 units

56. The meaning of the word trigonometry is three angles measure.
 (a) True (b) False (c) Can't determine (d) None of these

Ans: (a) True

57. Trigonometry is also applicable to obtuse angles triangles.
 (a) True (b) False (c) Can't determine (d) None of these

Ans: (b) False

58. If the length of the opposite side is 20 units and the length of the hypotenuse is 29 units then find cosec A?
 (a) $20 / 29$ (b) $29 / 21$ (c) $21 / 20$ (d) $29 / 20$

Ans: (d) $29 / 20$

59. If $\tan A = 89 / 17$, then $\cot A$ is ____
 (a) $89 / 17$ (b) $89 / 16$ (c) $17 / 89$ (d) $16 / 89$

Ans: (c) $17 / 89$

60. If $\cos C = 8 / 17$ and $\sec C = 17 / 8$, then 1 is the product of $\cos C$ and $\sec C$.
 (a) False (b) True (c) Can't determine (d) None of these

Ans: (b) True

61. The value of $\cos 0^\circ \cdot \cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 89^\circ \cos 90^\circ$ is
 (a) 1 (b) 0 (c) -1 (d) 5

Ans: (b) 0

62. If $x \tan 45^\circ \sin 30^\circ = \cos 30^\circ \tan 30^\circ$, then x is equal to
 (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{2}$ (d) None of these

Ans: (b) 1

63. If x and y are complementary angles, then
 (a) $\sec x = \operatorname{cosec} y$ (b) $\cos x = \cos y$ (c) $\tan x = \tan y$ (d) $\sin x = \sin y$

Ans: (a) $\sec x = \operatorname{cosec} y$

64. $\sin 2B = 2 \sin B$ is true when B is equal to
 (a) 0° (b) 30° (c) 60° (d) 90°

Ans: (a) 0°

65. If A and $(2A - 45^\circ)$ are acute angles such that $\sin A = \cos (2A - 45^\circ)$, then $\tan A$ is equal to

(a) 1 (b) $\sqrt{3}$ (c) 0 (d) None

Ans: (a) 1

66. If $y \sin 45^\circ \cos 45^\circ = \tan^2 45^\circ - \cos^2 30^\circ$, then y = ...
 (a) $1 / 2$ (b) -2 (c) 0 (d) $-1 / 2$

Ans: (a) $1 / 2$

67. If $\sin \theta + \sin^2 \theta = 1$, then $\cos^2 \theta + \cos^4 \theta = \dots$

- (a) 1 (b) 2 (c) 0 (d) -1

Ans: (a) 1

68. $5 \tan^2 A - 5 \sec^2 A + 1$ is equal to

- (a) -4 (b) 1 (c) -5 (d) 6

Ans: (a) -4

69. If $x = a \cos \theta$ and $y = b \sin \theta$, then $b^2 x^2 + a^2 y^2 =$

- (a) $a^2 b^2$ (b) $a^4 b^4$ (c) $b^2 + a^2$ (d) ab

Ans: (a) $a^2 b^2$

70. What is the minimum value of $\sin A$, $0 \leq A \leq 90^\circ$

- (a) -1 (b) 0 (c) 1 (d) $1/2$

Ans: (b) 0

71. What is the minimum value of $\cos \theta$, $0 \leq \theta \leq 90^\circ$

- (a) 0 (b) 1 (c) -1 (d) $1/2$

Ans: (a) 0

72. If $\cos 9A = \sin A$ and $9A < 90^\circ$, then the value of $\tan 5A$ is

- (a) 0 (b) 1 (c) $\sqrt{3}$ (d) None

Ans: (b) 1

73. If in $\triangle ABC$, $\angle C = 90^\circ$, then $\sin (A + B) =$

- (a) 1 (b) $1/2$ (c) 0 (d) $\sqrt{3}$

Ans: (a) 1

74. If $\sin A - \cos A = 0$, then the value of $\sin^4 A + \cos^4 A$ is

- (a) 1 (b) 2 (c) $1/2$ (d) $1/3$

Ans: (c) $1/2$

75. Ratios of sides of a right triangle with respect to its acute angles are known as

- (a) Trigonometric ratio of the angles (b) Trigonometry
(c) Trigonometric identities (d) None of these

Ans: (a) Trigonometric ratios of the angles

76. If $0^\circ < \theta < 90^\circ$, then $\sec \theta$ is

- (a) > 1 (b) < 1 (c) $= 1$ (d) 0

Ans: (a) > 1

77. If $\sin \theta = \sqrt{3} \cos \theta$, $0^\circ < \theta < 90^\circ$, then θ is equal to

- (a) 60° (b) 90° (c) 45° (d) 30°

Ans: (a) 60°

78. If $\cos (\alpha + \beta) = 0$, then $\sin (\alpha - \beta)$ can be reduced to

- (a) $\sin 2\alpha$ (b) $\sin \alpha$ (c) $\cos 2\beta$ (d) $\cos \beta$

Ans: (c) $\cos 2\beta$

79. If $\cos 9\alpha = \sin \alpha$ and $9\alpha < 90^\circ$, then the value of $\tan 5\alpha$ is
(a) $\sqrt{3}$ (b) 1 (c) 0 (d) None

Ans: (b) 1

80. $\sin(45^\circ + \theta) - \cos(45^\circ - \theta)$ is equal to
(a) $2 \sin \theta$ (b) $2 \cos \theta$ (c) 0 (d) 1

Ans: (c) 0

81. Which of the following is not a basic trigonometric ratio?
(a) Sine (b) Cosine (c) Tangent (d) Logarithm

Ans: (d) Logarithm

82. As θ increases from 0° to 90° , $\sin \theta$:
(a) Decreases (b) Increases (c) Stays the same (d) Is not defined

Ans: (b) Increases

83. Which angle gives $\sin \theta = 0$?
(a) 0° (b) 30° (c) 45° (d) 60°

Ans: (a) 0°

84. If $\cos \theta$ is positive and $\sin \theta$ is negative, θ is in which quadrant?
(a) I (b) II (c) III (d) IV

Ans: (d) IV

85. If A and B are complementary ($A + B = 90^\circ$), then $\sin A$ equals:
(a) $\cos A$ (b) $\cos B$ (c) $\tan B$ (d) 1

Ans: (b) $\cos B$

86. How many degrees are in π radians?
(a) 90° (b) 120° (c) 180° (d) 360°

Ans: (c) 180°

87. The value of $(\sin 30^\circ + \cos 30^\circ) - (\sin 60^\circ + \cos 60^\circ)$ is
(a) -1 (b) 0 (c) 1 (d) 2

Ans: (b) 0

88. The value of $\tan 30^\circ / \cot 60^\circ$ is
(a) $1/\sqrt{2}$ (b) $1/\sqrt{3}$ (c) 3 (d) 1

Ans: (d) 1

89. The value of $(\sin 45^\circ + \cos 45^\circ)$ is
(a) $1/\sqrt{2}$ (b) $\sqrt{2}$ (c) $\sqrt{3}/2$ (d) 1

Ans: (b) $\sqrt{2}$

90. Find the correct trigonometric identity.
(a) $\cos^2 \theta = 1 - \sin^2 \theta$ (b) $\cos^2 \theta = 1 + \sin^2 \theta$
(c) $\tan^2 \theta + \sec^2 \theta = 1$ (d) $\tan^2 \theta = \sec^2 \theta + 1$

Ans: (a) $\cos^2 \theta = 1 - \sin^2 \theta$

91. Find the correct trigonometric identity.

- (a) $\tan^2\theta = \sec^2\theta - 1$ (b) $\tan^2\theta + \sec^2\theta = 1$
(c) $\tan^2\theta - \sec^2\theta = 1$ (d) $\tan^2\theta = \sec^2\theta + 1$

Ans: (a) $\tan^2\theta = \sec^2\theta - 1$

92. Evaluate $(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta - \cot \theta)$.

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (b) 1

$$(\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta - \cot \theta) = \operatorname{cosec}^2\theta - \cot^2\theta = 1$$

93. Find the distance between $(\sin \theta, 0)$ and $(0, \cos \theta)$.

- (a) 0 (b) 1 (c) $4/3$ (d) $3/4$

Ans: (b) 1

$$\begin{aligned}\text{Distance between two points} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - \sin \theta)^2 + (\cos \theta - 0)^2} \\ &= \sqrt{\sin^2 \theta + \cos^2 \theta} = \sqrt{1} = 1\end{aligned}$$

94. _____ is drawn from the eye of an observer to the targeted object.

- (a) A parallel line (b) Line of sight
(c) Elevation line (d) Depression line

Ans: (b) Line of sight

95. What happens to the angle of elevation if the height of a tower, the distance between the tower and the observer is doubled?

- (a) Doubled (b) Halved (c) Tripled (d) Remains the same

Ans: (d) Remains the same

96. A slide should be built in a park that has to be set up at 10 meters height and also 10 meters away from the support. What should be the angle of elevation?

- (a) 45° (b) 30° (c) 60° (d) 90°

Ans: (a) 45°

97. The angle of depression is always a / an _____ angle.

- (a) Right (b) Obtuse (c) Complete (d) Acute

Ans: (d) Acute

98. Evaluate $(\operatorname{cosec}^2 \theta - \cot^2 \theta)^2 \cdot (\operatorname{cosec} \theta + \cot \theta)^2$.

- (a) 1 (b) 0 (c) $(\operatorname{cosec}^2 \theta - \cot^2 \theta)^2$ (d) $(\operatorname{cosec} \theta + \cot \theta)^2$

Ans: (d) $(\operatorname{cosec} \theta + \cot \theta)^2$

99. Evaluate $\sqrt{(1 - \sin A) / (1 + \sin A)}$

- (a) $\cos A + \tan A$ (b) $\cos A - \tan A$
(c) $\tan A - \cot A$ (d) $\sec A + \tan A$

Ans: (d) $\sec A + \tan A$

$$\sqrt{\frac{1-\sin A}{1+\sin A}} = \sqrt{\frac{1-\sin A}{1+\sin A}} \cdot \sqrt{\frac{1-\sin A}{1-\sin A}} = \frac{\sqrt{(1-\sin A)^2}}{\sqrt{1-\sin^2 A}} = \frac{1-\sin A}{\sqrt{\cos^2 A}}$$

$$(\because \sin^2 A + \cos^2 A = 1) \therefore = \frac{1-\sin A}{\cos A} = \sec A - \tan A$$

100. $(\sin A + \cos A)^2$ is equal to _____

- (a) $1 + 2\sin A \cos A$ (b) $1 - 2\sin A \cos A$
(c) $2\sin A \cos A - 1$ (d) $2\sin A \cos A + 1$

Ans: (a) $1 + 2\sin A \cos A$

$$(\sin A + \cos A)^2 = \sin^2 A + \cos^2 A + 2\sin A \cos A = 1 + 2\sin A \cos A$$